

From policy question to reproducible brief

Coding Lesson 4 · Four questions, one workflow

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Lesson map

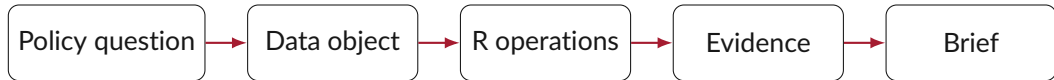
The investigation so far

The R vocabulary

From script to report

Prepare the lab

The Math Camp workflow



- ▶ Lesson 1 established the division of work between analyst, R, and agent.
- ▶ Lesson 2 constructed a documented analysis table.
- ▶ Lesson 3 moved from observations to plots and descriptive claims.
- ▶ Lesson 4 combines the work in a report another reader can render.

Four questions remain available

1. **Health.** How is income associated with under-five mortality in 2022, and how does the pattern differ by income group?
2. **Gender.** Where is girls' secondary enrollment associated with lower adolescent fertility, and where does the pattern depart from the overall comparison?
3. **Access.** Which economies closed the combined electricity and internet access gap fastest from 2000 to 2022?
4. **Climate.** Which economies expanded renewable electricity, reduced carbon intensity, and ended with higher real income from 2000 to 2021?

The final student artifact

One rendered Quarto brief contains:

1. the policy question;
2. source, unit, sample, and period;
3. one table;
4. one figure;
5. one supported finding;
6. one limitation; and
7. one AI-use and verification note.

The artifact is optional and ungraded. Its value is that every result remains connected to code and evidence.

Two hours before Lab 4

Minutes	Part	What we will accomplish
00-45	Function recap	Read, inspect, transform, summarise, and plot.
45-51	Checkpoint 1	Choose a defensible function sequence.
51-61	Data design	Distinguish cross-sections from endpoint comparisons.
61-71	Break	Ten minutes away from the screen.
71-104	Quarto	Build and render one complete brief.
104-116	Checkpoints 2-3	Diagnose a broken evidence chain and plan a track.
116-120	Lab handoff	Open the four starter files and choose one.

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A compact map of the functions

Task	Core functions	Question answered
Start	<code>library()</code> , <code>read_csv()</code>	What object did we load?
Inspect	<code>glimpse()</code> , <code>names()</code> , <code>count()</code>	What does a row represent?
Choose rows/columns	<code>filter()</code> , <code>select()</code>	Which observations and variables belong?
Construct	<code>mutate()</code> , <code>arrange()</code>	What new measure or ordering is useful?
Aggregate	<code>group_by()</code> , <code>summarise()</code>	What statistic belongs to each group?
Visualize	<code>ggplot()</code> , <code>aes()</code> , <code>geom_*()</code> , <code>labs()</code>	What comparison should the reader see?

Install once; attach each session

```
# Run only if tidyverse is absent.  
install.packages("tidyverse")  
  
# Run at the start of each session that uses it.  
library(tidyverse)
```

Installation puts a package on the computer. `library()` makes its functions available in the current R session.

The file format determines the reader

```
health <- read_csv("data/lesson-4/health-2022.csv")  
  
# Related readers:  
# readxl::read_excel("file.xlsx")  
# object <- readRDS("object.rds")
```

- ▶ CSV: rectangular text data.
- ▶ Excel: one or more worksheets.
- ▶ RDS: one R object saved with its classes and attributes.

Inspect before transforming

```
glimpse(health)
names(health)
count(health, income_level_current)
summary(select(health, gdp_per_capita_ppp,
                 under5_mortality))
n_distinct(health$iso3c)
```

The first reading should establish the row unit, the key, the variables, their types, and missing coverage.

Focus on economies above \$10,000 per person

```
health_focus <- health |>
  select(country, region, income_level_current,
         gdp_per_capita_ppp, under5_mortality) |>
  filter(gdp_per_capita_ppp > 10000)
```

`select()` retains the variables needed for the comparison. `filter()` defines the analytical sample: the tables and figures that follow describe economies with GDP per capita, PPP, above \$10,000 in 2022.

Construct and order variables

```
health_focus <- health_focus |>  
  mutate(log_income = log(gdp_per_capita_ppp)) |>  
  arrange(desc(under5_mortality))
```

`mutate()` preserves the row unit while adding or changing columns. `arrange()` changes display order; it does not change the sample.

Summarise within groups

```
health_table <- health_focus |>
  group_by(income_level_current) |>
  summarise(
    economies = n(),
    min_mortality = min(under5_mortality),
    average_mortality = mean(under5_mortality),
    median_mortality = median(under5_mortality),
    max_mortality = max(under5_mortality),
    .groups = "drop"
  )
```

The output has one row per income group. The count and range describe the sample; the mean and median describe its center.

Map variables to a figure

```
ggplot(health_focus,  
      aes(x = gdp_per_capita_ppp,  
          y = under5_mortality,  
          color = income_level_current)) +  
geom_point(alpha = 0.75) +  
scale_x_log10() +  
labs(x = "GDP per capita, PPP (log scale)",  
     y = "Deaths per 1,000 live births",  
     color = "Current income group") +  
theme_minimal()
```

Checkpoint 1: choose the sequence

You need the three economies in each income group with the highest under-five mortality in 2022.

1. What should one row of the final object represent?
2. Which functions would you use, and in what order?
3. What information would disappear if you summarised too early?

2 minutes discuss → 2 minutes share

Checkpoint 1: resolution

One defensible sequence is:

```
filter(year == 2022) → group_by(income group) → slice_max(mortality, n =  
3)
```

The resulting row remains an economy. A group-level `summarise()` would remove economy names and prevent the requested ranking.

Cross-section or endpoint comparison?

Design	Unit	Appropriate question
2022 cross-section	economy in 2022	How do economies differ at one date?
Fixed endpoints	economy with the same two years observed	How much did each economy's measure change between two dates?
Country-year panel	economy-year	How do levels and changes evolve over time?

The health and gender files are cross-sections. The access and climate files use fixed endpoints.

Break

10 minutes

When we return, we will place the health analysis in a Quarto document and render it from a clean session.

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Why use a rendered report?

A script can compute a result. A policy brief must also tell the reader:

- ▶ the question and decision context;
- ▶ the source, sample, unit, and period;
- ▶ how the table and figure were produced;
- ▶ what the evidence supports; and
- ▶ what remains uncertain.

Quarto keeps the prose, code, and generated output in one document.

R Markdown and Quarto

	R Markdown	Quarto
Source file	.Rmd	.qmd
Code chunks	R and other engines	R and other engines
Typical outputs	HTML, PDF, Word	HTML, PDF, Word, slides, books
Course use	Existing reports remain valid	New Math Camp briefs use Quarto

Both combine narrative and code. We use Quarto because one syntax supports the student book, slides, and standalone briefs.

Anatomy of a Quarto file

```
---  
title: "Income and under-five mortality"  
format:  
  html:  
    toc: true  
    embed-resources: true  
---  
  
## Data and sample  
  
```{r}  
library(tidyverse)
health <- read_csv("data/lesson-4/health-2022.csv")
```
```

The YAML controls the document. Markdown supplies prose and headings. R chunks compute the evidence.

Live demonstration: build the health brief

Open `materials/lesson-4-demo.qmd` beside RStudio.

1. Read the question and sample statement.
2. Run the import and inspection chunk.
3. Run the grouped table.
4. Run the figure.
5. Compare the prose with the output.
6. Click **Render**.

We will deliberately change one object name and render again to observe how a broken dependency appears.

Rendering checks the document from the beginning

When you click Render, Quarto starts a fresh R session and executes the chunks in order.

- ▶ An object created only in the Console is unavailable.
- ▶ A missing package or file path becomes visible.
- ▶ A table or figure must be generated by code in the document.
- ▶ A successful render produces a portable record of the analysis.

Checkpoint 2: repair the evidence chain

A draft brief contains:

- ▶ a table restricted to 2022;
- ▶ a figure pooling all years from 2000 to 2022; and
- ▶ the sentence, “Higher income reduces under-five mortality.”

Identify two problems and propose the smallest defensible revision to the table, figure, or sentence.

2 minutes discuss → 2 minutes share

Checkpoint 2: resolution

1. The table and figure use different periods and samples. A reader cannot treat them as evidence for the same comparison without explanation.
2. The sentence is causal. The cross-sectional comparison supports an association, not the effect of an income intervention.

Smallest revision: align both outputs to 2022 and describe the observed association and its variation.

The agent enters after the first interpretation

Save the table, figure, finding, and limitation before opening Codex. **The read-only request asks Codex to:**

1. reproduce the table from the stated sample;
2. inspect the variables, units, and figure mappings;
3. challenge one consequential statement with exact evidence; and
4. end with accept, revise, or reject and at most one revision.

The complete prompt is in the Lab 4 handout. Independently verify one material statement in R.

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Four prepared files

| Track | Unit | Rows | Main comparison |
|---------|------------------|------|--|
| Health | economy, 2022 | 188 | income and under-five mortality |
| Gender | economy, 2022 | 140 | girls' enrollment and adolescent fertility |
| Access | economy, 2000–22 | 174 | electricity and internet percentage-point changes |
| Climate | economy, 2000–21 | 172 | renewable, carbon-intensity, and income endpoint changes |

The prepared files hold the unit and comparison fixed so the lab can focus on analysis and communication.

Checkpoint 3: plan one track

Choose a question. Before writing code, state:

1. the row unit and period;
2. the variables required;
3. the function sequence for the table;
4. the variables mapped to the figure;
5. one claim the evidence could support; and
6. one claim it could not support.

2 minutes discuss → 2 minutes share

Lab 4 route

| Minutes | Work | Result |
|---------|-------------------|--|
| 00-07 | Choose and plan | Unit, variables, and function sequence. |
| 07-20 | Construct table | One traceable briefing table. |
| 20-33 | Construct figure | One labeled policy comparison. |
| 33-40 | Interpret | One finding and one limitation in your words. |
| 40-50 | Agent review | Read-only reproduction and claim challenge. |
| 50-56 | Independent check | Accept, revise, or reject one agent statement. |
| 56-60 | Render | Standalone HTML brief. |

Where the materials are

- ▶ Lesson code: `code/lesson-4-walkthrough.R`
- ▶ Complete demonstration: `materials/lesson-4-demo.qmd`
- ▶ Student starters: `materials/lab-4/`
- ▶ Prepared data: `data/lesson-4/`
- ▶ Lab instructions: `labs/lab-4/handout/lab-4-handout.pdf`

Open the project in RStudio, place the slides beside the walkthrough, and run one section at a time.

Before tomorrow

Open the four starter files and choose a track. Confirm that:

1. the corresponding CSV opens;
2. you can run its import chunk;
3. you understand the row unit and period; and
4. the Quarto Render button is available in RStudio.

No model is required. Lab 4 produces a descriptive table, a figure, an interpretation, and a reproducible report.